UX Primer for Frontend Devs

An abridged field guide to the principles behind good interfaces, with examples from fomodoro. Prepared November 2026; sources and evidence labels added November 2026.

How to read this. Each idea gets a plain-English definition and an example from our app (in green). You do not need to memorize any of it. The goal is to recognize these patterns when you review a screen or a PR and to have a name for what feels off. Numbers in brackets, like [1], point to the numbered sources at the end. Sentences starting "We recommend" are our own judgment, not findings.

Evidence labels. Each concept carries a tag saying how much weight it can bear.

STRONG Replicated empirical research, or a formal standard (such as WCAG).

MODERATE Some empirical support, but limited, dated, proprietary, or contested in scope.

HEURISTIC Expert or practitioner guidance and observation; not experimentally established. Official platform guidance (Apple HIG) gets this label too: it is the authority on that platform's conventions, but it is not itself evidence.

WEAK Tested empirically, but current evidence does not support the popular claim.

A Strong finding in a lab does not automatically make our app-specific application of it Strong. Where that gap matters, the text says so.

1. Nielsen's 10 usability heuristics HEURISTIC

Jakob Nielsen and Rolf Molich proposed an early set of usability heuristics in 1990 [1]. In 1994 Nielsen alone revised them into the ten below, derived from a factor analysis of 249 usability problems [2]; NN/g still publishes them [3]. They are the most widely used checklist for reviewing an interface. The heuristics themselves are expert rules of thumb. *Heuristic evaluation* as a method has been studied empirically MODERATE: single evaluators miss many problems, so several are needed, and different evaluators find different problems [1, 4].

Heuristic	What it means	In fomodoro
1. Visibility of system status	Always show what's happening, promptly.	Big countdown, block name, cycle dots, time in the tab title.
2. Match the real world	Use the user's words and familiar concepts, not system jargon.	"Short break", not "Interval type 2".
3. User control and freedom	Give a clear way out of mistakes: undo, cancel, back.	Pause, Skip, Reset. Reset asks nothing because it's cheap to redo.

4. Consistency and standards	Same words and actions mean the same thing everywhere; follow platform conventions.	Space toggles start/pause, just like media players.
5. Error prevention	Better than good error messages: design so the error can't happen.	Steppers instead of free-typed minutes, so "2500" is impossible.
6. Recognition over recall	Show options and state; don't make people remember them.	"Next: short break, 5 min" so you never count blocks.
7. Flexibility and efficiency	Shortcuts for experts that novices can ignore.	Keyboard shortcuts; the mini timer window.
8. Aesthetic, minimalist design	Every extra element competes with the important ones.	One primary button; settings behind one icon.
9. Help users recover from errors	Plain-language errors that say what happened and what to do.	"Code expired. Send a new one." with the button right there.
10. Help and documentation	Ideally unnecessary; when needed, short and in context.	The iPhone install sheet appears exactly when you need it.

2. The "laws of UX"

Jon Yablonski's *Laws of UX* (book and lawsofux.com) popularized these as a designer's vocabulary [5]. Despite the shared label, they are a mixed bag: some are robust laboratory findings, some are single studies, and some are practitioner sayings. The tags matter here more than anywhere else in this primer.

Hick's law (Hick-Hyman law): more choices, slower decisions STRONG IN THE LAB MODERATE AS A UI RULE

In choice-reaction experiments, the time to respond grows roughly with the logarithm of the number of equally likely alternatives [6, 7]. That result is well replicated, but it was measured with practiced stimulus-response pairs (a light comes on, press its key). Real interfaces add visual search, reading, and unfamiliar labels, and for an unfamiliar menu people often scan item by item, which grows closer to linearly [8, 9]. So treat Hick's law as a direction, not a formula. We recommend reducing options at moments that need speed and grouping or hiding the rest. It does *not* mean "fewer features"; it means fewer choices *at once*.

The timer screen has one primary action. All configuration lives in Settings.

Fitts's law: big and close is fast STRONG

The time to hit a target depends on its distance and size: bigger and nearer is faster and less error-prone [10]. This is one of the best-replicated models in HCI, including for mice and touch [11]. A derived rule of thumb: screen edges and corners act as very large targets on desktop because the cursor stops there.

Large round Pause button near the thumb zone; Reset and Skip smaller and away from it so they're not tapped by accident.

Law	The idea and its evidence	Practical use
Jakob's law HEURISTIC	Users spend most of their time on *other* sites and expect yours to work the same way. An observation from Nielsen, not an experimental result [12].	Don't reinvent settings screens, sign-in, or gestures.
Doherty threshold HEURISTIC	An IBM report argued that productivity rises sharply when system response drops below about a second [13]; the popular "400 ms" figure is attributed to it but we could not confirm the number in the primary text. Dated (1982, mainframe users) and vendor-authored. That faster feedback feels better is broadly supported by older response-time guidance [14, 15].	Optimistic UI: show the change instantly, sync after.
Miller's law STRONG	Working memory is limited [16]. The famous "7 ± 2" is overstated; a widely cited review puts the limit nearer 4 chunks [17].	Chunk information. Never use it to justify "max 7 menu items".
Tesler's law HEURISTIC	Every system has some complexity that cannot be removed; the question is whether the system or the user carries it. A practitioner principle from Larry Tesler [18].	We handle iOS push quirks so you just see "Break time".
Goal-gradient effect MODERATE	Effort increases as a goal gets closer. Shown in rats [19] and in field studies of coffee-card rewards [20]; boundary conditions are still debated.	The progress ring and cycle dots make the finish line visible.
Zeigarnik effect WEAK	The claim that unfinished tasks are remembered better [21] did not hold up in a 2025 meta-analysis [22]. The same analysis did support the related *Ovsiankina effect*: people tend to resume interrupted tasks [23] MODERATE.	Don't rely on "it'll nag at them". Make resuming easy instead.
Peak-end rule STRONG	When people rate a past experience, they weight its most intense moment and its ending heavily and largely ignore its duration [24, 25]; a 2022 meta-analysis supports this [26]. Applying it to app moments is our inference.	Make the "block done" moment feel good: a pleasant chime, not a klaxon.
Aesthetic-usability effect MODERATE	People rate attractive designs as easier to use [27, 28]. The studies measured *perceived* usability; later work shows the link varies with context [29].	Polish matters, but it can hide real usability problems in testing.
Von Restorff (isolation) effect STRONG	An item that differs from its neighbours is remembered better [30, 31]. That's a memory finding; using it to draw attention to a button is an extension.	Make exactly one thing visually distinct: the primary action.

Gestalt: proximity, similarity, common region STRONG	Things close together, alike, or inside one boundary are seen as a group [32, 33, 34].	We recommend using spacing as the first grouping tool, before borders and lines.

3. Don Norman's design principles HEURISTIC

From *The Design of Everyday Things* (1988 as *The Psychology of Everyday Things*; revised 2013) [35], the foundation much of the above builds on. These are an influential conceptual framework, not experimental results. Norman joined Apple in 1993 and is widely credited with popularizing "user experience"; an Apple CHI '95 overview he co-wrote is one of the earliest published uses of the term [36].

- **Affordances and signifiers.** An affordance is what an object lets you do; a signifier is the visible clue [35, 37]. A flat text label that's secretly a button has the affordance but no signifier.
- **Feedback.** Every action gets an immediate, visible response. Pressing Start instantly changes the button to Pause and starts the ring moving.
- **Mapping.** Controls should relate obviously to what they affect. A stepper's minus on the left and plus on the right maps to "less" and "more".
- **Constraints.** Limit what can be done so the wrong thing is hard to do (pairs with Nielsen #5).
- **Conceptual model.** Users build a mental model of how the system works; the design should make the right one easy to form. "A cycle is 4 focus blocks with breaks, then a long break."

4. Progressive disclosure HEURISTIC

Definition: show only what most people need for the current task, and make the rest available on request. It manages complexity by *sequencing* it, not by removing it. The idea dates to 1980s interface work; an early study found that beginners blocked from advanced functions ("training wheels") learned faster [38] MODERATE. Nielsen's 2006 article is the standard practitioner summary [39].

Common forms

- **On-demand:** secondary options behind a menu, "Advanced" section, or settings icon. Durations, sounds, 2FA.
- **Staged (step-by-step):** a flow split into screens, each with one job. Sign-in: email first, then the code screen.
- **Contextual (just-in-time):** information that appears at the moment it becomes relevant. "Install the app to get notifications" appears only when you turn notifications on in iPhone Safari.
- **Expandable details:** accordions, "Show more", tooltips for explanations.

How to get it right

- **Decide what goes first using data on use.** Nielsen recommends frequency-of-use data plus observation for existing products, and task analysis for new ones [39].
- **Keep it shallow.** Nielsen observes that designs with more than two levels of disclosure tend to have low usability because people get lost [39]. If you need three, restructure.
- **Make the path visible.** The hidden content needs a clear signifier ("More options", a gear icon) or it effectively doesn't exist.

- **Don't hide what people need to finish the task**, especially errors, prices, or consequences. That crosses into deceptive design (Section 7).

> **Same idea, applied to Claude.** This is our analogy, not a research claim. Our context-management rules (short AGENTS.md, rules that load only for matching files, skills that load on demand) are progressive disclosure for an AI: show the essential first, the rest on request.

5. Apple Human Interface Guidelines (HIG) HEURISTIC

Apple's official design guidance for its platforms (developer.apple.com/design/human-interface-guidelines). We recommend following it even in a web app, because it makes the app feel native on iPhone and Mac. Two framings are current. The HIG's own "Design principles" page lists eight: purpose, agency, responsibility, familiarity, flexibility, simplicity, craft, and delight [40]. Separately, Apple introduced its 2025 Liquid Glass design system around **hierarchy** (controls float in a layer above content, keeping content primary), **harmony** (shapes follow the device's hardware and the curve of a finger), and **consistency** (one design language across platforms) [41].

Guideline	What Apple says, and our rule of thumb
Touch targets MODERATE	Apple lists 44 × 44 pt as the default control size on iOS (28 × 28 pt as the minimum) [42]. Thumb-tapping studies support targets of roughly 7–10 mm [43]. We treat 44 pt as the target.
Safe areas	Keep content clear of the notch, Dynamic Island, and home indicator (CSS `env(safe-area-inset-*)`).
Dynamic Type	Text should scale with the user's text-size setting. On the web, size text in `rem` so it follows the browser's text size. iOS Safari follows the system Dynamic Type setting only if the page opts in (for example the WebKit-only `font: -apple-system-body`).
Dark Mode	Support it with real dark palettes, not just inverted colors.
Standard gestures	Don't override system gestures (edge swipe for back, swipe up for home).
Notifications	Concise and actionable. We recommend never using them for marketing.
Motion	Purposeful and brief; honor "Reduce Motion" (`prefers-reduced-motion`).

Google's equivalent is Material Design 3 (m3.material.io). Worth skimming for its component guidance even if we don't use it.

6. WCAG 2.2 (accessibility) STRONG (STANDARD)

The Web Content Accessibility Guidelines from the W3C. Version 2.2 became a W3C Recommendation on 5 October 2023; the current edition is dated 12 December 2024 [44]. WCAG 3 is still a working draft. Levels are A, AA, and AAA; **AA is the usual target**. Laws and regulations typically reference Level AA, though often of WCAG 2.0 or 2.1 rather than 2.2 (for example the 2024 US ADA Title II rule and the EU's EN 301 549) [45, 46]. Everything hangs off four principles, remembered as **POUR**:

- **Perceivable:** content can be seen or heard (text alternatives, contrast, captions).
- **Operable:** everything works by keyboard, there's enough time, nothing triggers seizures.
- **Understandable:** readable text, predictable behavior, help with input errors.
- **Robust:** works with assistive technology, which mostly means correct semantic HTML.

The AA criteria you'll hit most

Criterion	Requirement
Contrast (1.4.3)	4.5:1 for normal text, 3:1 for large text (18 pt, or 14 pt bold).
Non-text contrast (1.4.11)	3:1 for UI component boundaries, states, and meaningful graphics.
Use of color (1.4.1, Level A)	Color can't be the only way information is conveyed. Why "Focus" stays as text on the tinted screen.
Keyboard (2.1.1, A) and Focus visible (2.4.7)	Everything works without a mouse, and the keyboard focus is visible.
Status messages (4.1.3)	Screen readers announce changes without moving focus (ARIA live regions). "Short break started."
Reflow (1.4.10)	Works at 320 CSS px wide (equivalent to 400% zoom) without two-direction scrolling.

New in 2.2

- **Focus not obscured (2.4.11, AA):** the focused element can't be entirely hidden behind sticky headers or banners.
- **Target size, minimum (2.5.8, AA):** at least 24 × 24 CSS px, or enough spacing around smaller targets. Apple's 44 pt is stricter; we follow Apple.
- **Dragging movements (2.5.7, AA):** anything draggable also needs a non-drag alternative.
- **Accessible authentication, minimum (3.3.8, AA):** no memory or transcription test to sign in unless there's an alternative or help, such as paste and password managers. AA still allows object-recognition checks. Our 6-digit code field must allow paste and use `autocomplete="one-time-code"` so iOS offers the code from Mail.

- **Redundant entry (3.3.7, A)** and **consistent help (3.2.6, A):** don't make people retype what they already entered; keep help in the same place.

> **The first rule of ARIA** (W3C guidance [47]): use a native HTML element (`<button>`, `<input>`, `<dialog>`) before reaching for ARIA attributes. Native elements come with keyboard and screen-reader behavior built in; a `<div onclick>` does not.

7. Modern frontend practice: the rest of the toolkit

- **Performance is UX.** MODERATE Google's Core Web Vitals measure it, at the 75th percentile of page loads: **LCP** (main content visible, ≤ 2.5 s), **INP** (response to taps and clicks, ≤ 200 ms; it replaced FID in 2024), **CLS** (layout doesn't jump, ≤ 0.1) [48]. Google set these from perception research plus what real sites can achieve; the CLS cut-off came from Google's own testing [49]. Tabular digits on the timer stop the numbers jittering every second, a visual-stability nicety.
- **Design systems and tokens.** HEURISTIC Colors, spacing, and type defined once as named variables (`--color-focus`, `--space-4`) and reused everywhere. Consistency becomes much easier and dark mode becomes a token swap.
- **Mobile-first, responsive.** HEURISTIC Design the smallest screen first, then add [50]. Container queries let components adapt to their own space, not just the window.
- **Respect user preferences.** HEURISTIC `prefers-color-scheme`, `prefers-reduced-motion` [51], text size, Focus modes.
- **Visual hierarchy through restraint.** HEURISTIC Size, weight, color, and space. In our experience most "it looks amateur" problems are spacing and too many competing weights. We recommend *Refactoring UI* as a practical book on this [52].
- **Avoid deceptive patterns** MODERATE (Harry Brignull coined "dark patterns" in 2010 and now uses "deceptive patterns" [53]): fake urgency, hidden costs, guilt-shaming opt-outs, roach-motel cancellation. They are common on real sites [54] and experiments show they change what people agree to [55]. They are increasingly regulated: the EU Digital Services Act bans them on online platforms [56], and the US FTC has published a staff report on them [57].
- **Inclusive design.** HEURISTIC One of Microsoft's three inclusive-design principles is to solve for one person and extend the benefit to many [58]. Captions help deaf users *and* people in loud rooms; big targets help motor impairments *and* people walking.
- **Test with real use.** MODERATE Nielsen and Landauer's model suggests about five users find most of the problems a given test can surface [59, 60], but later studies found five users sometimes found as few as about half [61]. For a single-user app, we recommend dogfooding plus a short note whenever something annoys you.

8. Further reading, in order (our recommendations)

- **Steve Krug, *Don't Make Me Think, Revisited*** [62]: the fastest, most readable intro. Start here.

- **Nielsen Norman Group (nngroup.com/articles)**: free articles on everything above. Many draw on NN/g's own studies, which are useful but mostly not peer-reviewed.
- **lawsofux.com** [5]: one page per law, with sources. Read the sources: the laws vary a lot in strength.
- **Wathan and Schoger, *Refactoring UI*** [52]: practical visual design for developers.
- **Don Norman, *The Design of Everyday Things*** [35]: the foundational theory.
- **Apple HIG** and **WCAG 2.2 Quick Reference (w3.org/WAI/WCAG22/quickref)**: reference, not cover-to-cover reading.
- **web.dev**: Google's guides on performance, PWAs, and accessibility.

Sources and evidence strength

Evidence summary

Concept	Label	Why
Nielsen's heuristics	HEURISTIC	Expert-derived rules; evaluation method studied, with known evaluator effect [1, 2, 4].
Hick-Hyman law	STRONG / MODERATE	Robust in choice-reaction tasks; weaker predictor for menus and search [6, 7, 8, 9].
Fitts's law	STRONG	Replicated for decades across pointing devices [10, 11].
Jakob's law, Tesler's law	HEURISTIC	Practitioner observations [12, 18].
Doherty threshold	HEURISTIC	Single 1982 vendor report; 400 ms figure not confirmed in primary text [13].
Working-memory limit	STRONG	Limit well established; "7 ± 2" superseded by ~4 chunks [16, 17].
Goal-gradient	MODERATE	Animal and field evidence; boundary conditions debated [19, 20].
Zeigarnik / Ovsiankina	WEAK / MODERATE	Meta-analysis: no reliable memory advantage; resumption tendency supported [22].
Peak-end rule	STRONG	Original experiments plus 2022 meta-analysis [24, 25, 26].
Aesthetic-usability	MODERATE	Correlational, perceived (not measured) usability; context-dependent [27, 28, 29].
Von Restorff, Gestalt	STRONG	Replicated perception and memory findings; UI use is extrapolation [30, 31, 32, 33, 34].
Norman's principles; progressive disclosure	HEURISTIC	Influential frameworks; limited direct testing [35, 38, 39].
Apple HIG	HEURISTIC	Authoritative platform guidance [40, 41, 42].
44 pt targets	MODERATE	Apple default, consistent with thumb-target studies [42, 43].
WCAG 2.2	STRONG	Formal W3C standard [44].
Core Web Vitals	MODERATE	Google-defined from perception research plus achievability [48, 49].
Deceptive patterns	MODERATE	Prevalence and effect studies; regulatory facts [54, 55, 56, 57].
Five-user testing	MODERATE	Model-based; contested by larger-sample studies [59, 60, 61].

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